FAST NUCES, Islamabad Campus

F26-060 · BS (DS) · FYP 2026

Granite

AI-agent
investigation workbench

Find why AI agents fail.

Registered: A Self-Healing Multi-Agent System
for Semantic Error Diagnosis, Verified Repair and Recovery

PROJECT GOALS

- Detect silent failures
- Locate their source
- Verify proposed fixes

For engineers & QA teams

PROJECT TEAM

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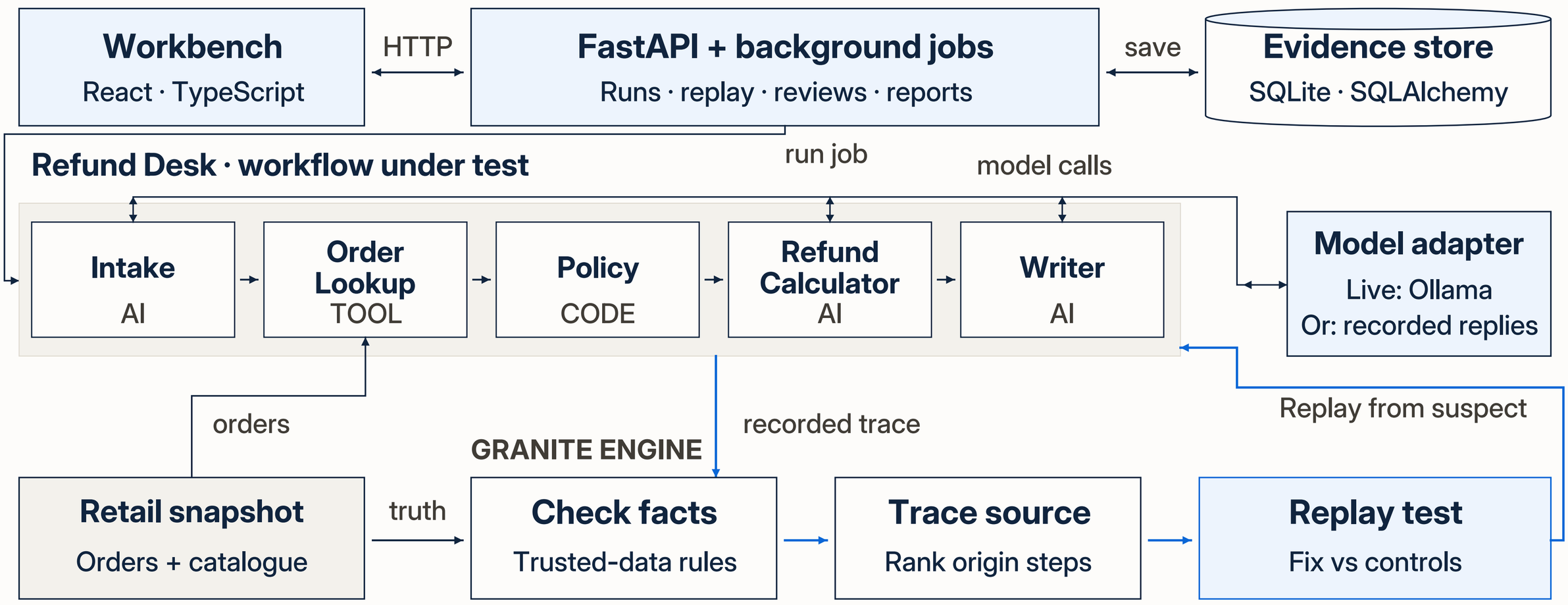
SUPERVISOR

Dr. Adil Majeed

01 / SYSTEM ARCHITECTURE · IMPLEMENTED

One investigation. Every layer connected.

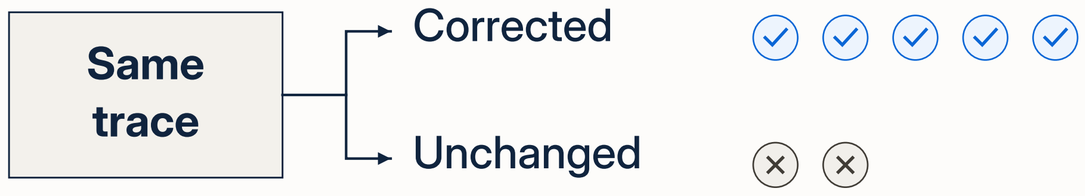
A run can finish without an error—and still be wrong.



API saves verdicts, evidence and human reviews · supplied repairs only

02 / CONTROLLED REPLAY

Compare a correction with controls.



Recorded case · supplied repair rule

EARLY FEASIBILITY

10/10
faulty runs detected

9/10
origins verified

10 tasks · 1 fault · 1 small model
Four tasks not blind

03 / VISION & TENTATIVE FYP-1 PLAN

Turn an investigation into a lasting test.

BUILT
1 · Investigate
Refund Desk workbench
Checks · trace · replay

PLANNED
2 · Guide repairs
One-click investigation
Check-derived fixes

PLANNED
3 · Extend
Retail Agent · imports
Regression-test export

04 / TOOLS & TECHNOLOGIES

ENGINE	API	INTERFACE	MODELS
Python	**FastAPI**	**React · TypeScript**	**Ollama**
Checks & tracing	Runs & replay jobs	Inspect & review	Local model calls